

```

{
  "generation_protocol": {
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    "seed": 20260921,
    "sources": [
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        "source_frames": 974,
        "branch_frame": 486,
        "branch_completion": 0.5,
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        "initial_heights": [
          0.790903627872467,
          0.7920903563499451
        ],
        "mug_addresses": [
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          7
        ],
        "split": "dev",
        "source_hashes": {
          "scene_assembled.xml": "0c126dc732bfa0c2e67823942f10f79607cea2917f95abe9911c71cb6335d63",
          "scene_qpos.npy": "681134f39a5d9b072f27f2bca566dcfa49ab88fa4b81a90db2f642bd66629af0",
          "states_actions.bin": "8287fa068ec796635d9e2c494011da256851c596ff0e5f3d37000815bcb622e",
          "randomization.json": "2b64f59c34654455bcc22ff6e0868f0847b0dec82150dd106583902d87f21d59"
        }
      },
      {
        "episode_id": "episode_019e3c90-fb67-765e-8e8e-12c56ddda3ba",
        "source_frames": 759,
        "branch_frame": 379,
        "branch_completion": 0.5,
        "source_final_completion": 1.0,
        "initial_heights": [
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          0.8087994456291199
        ],
        "mug_addresses": [
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          "scene_assembled.xml": "bcfde7524898d9af1ccb75ba3dc4fcbc02a28ec0abb947616b9a85edf9190694",
          "scene_qpos.npy": "26e71427fc7c6e563af0b30a27a0e7e5b5e8fee206ede7447eb66ef468e8a65d",
          "states_actions.bin": "311378a0e0cebfb57eedb13cd933c9b16eaf1d065b9043ff5086a0098b2bf49",
          "randomization.json": "a53536a56e62d182c79c49689436c10d8d3411501b02ce416b7b3bd65635b2eb"
        }
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        "source_frames": 734,
        "branch_frame": 366,
        "branch_completion": 0.5,
        "source_final_completion": 1.0,
        "initial_heights": [
          0.7958173751831055,
          0.8027102947235107
        ],
        "mug_addresses": [
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        "split": "test",
        "source_hashes": {
          "scene_assembled.xml": "1535462a4a2756e1a40fe8521c5437666a3d0c9a5ceb1b876865638b8a113973",
          "scene_qpos.npy": "dbf587fc1fe97fe0d9a66671c251cc67be2eb241fb45f7739fe7dd6801172324",
          "states_actions.bin": "5c8344d8b894dd54d62fe0f67e55000a2a5c3ec8f40e2c6da500f28b8674f6c9",
          "randomization.json": "c3c6b70907e982311cf12be72b7d2668c82cbaaa51fad73e1518d5e1e12db28d"
        }
      }
    ]
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}

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{
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  "branch_completion": 0.5,
  "source_final_completion": 1.0,
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    "scene_qpos.npy": "ef89eef1d0701163293b7dfa5949cff3d1fbc83efead01615ebe3dc0e8fe342e",
    "states_actions.bin": "840698fdb1fce3dc64baaf8349cd3101424f88ec305c9c8993090dfef4fc1a14",
    "randomization.json": "de8dede602cff8e73d8e9489e9a28573dd38dc18ffa369d635127a07e8de98ef"
  }
},
{
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  "source_frames": 828,
  "branch_frame": 413,
  "branch_completion": 0.5,
  "source_final_completion": 1.0,
  "initial_heights": [
    0.8089275360107422,
    0.808255532455444
  ],
  "mug_addresses": [
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    7
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  "source_hashes": {
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    "scene_qpos.npy": "00d4cc5babee09e7681da3eef9c6422ab310a61c55eb3432942fe2194bb550dd",
    "states_actions.bin": "925d141a20c67c2ce4c2e3c990187cb5c7fcc3af4990508f0e409d8ea0073bd5",
    "randomization.json": "76a76d91e7addb71f72fe66d67e13ef19fd276658be09d18428f6577e1bae95c"
  }
},
{
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  "source_frames": 655,
  "branch_frame": 327,
  "branch_completion": 0.5,
  "source_final_completion": 1.0,
  "initial_heights": [
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    0.8042730689048767
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  "mug_addresses": [
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    7
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  "split": "test",
  "source_hashes": {
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    "states_actions.bin": "6e91c93d4c08f01ee9e305ff2ae4e0eea460d796b900f1ac7c9c6254118eb5c3",
    "randomization.json": "1d52fd61e1a62b8623106d45ffb6d244b40b1bd8807395d2d2c482cdc0be0a6b"
  }
}
],
"screen": [
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    "eligible": false,
    "num_mugs": 1,
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    "branch_completion": 0.5
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    "eligible": true,
    "num_mugs": 2,
    "branch_completion": 0.5
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    "branch_completion": 0.5
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    "num_mugs": 2,
    "branch_completion": 0.5
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"modes": [
  "replay_control",
  "hold",
  "random_walk"
],
"branch_fraction": 0.5,
"fps": 30,
"physics_dt": 0.002,
"duration": "source duration plus 2 seconds, equal for every branch of a source",
"physics_clock": "alternate 16/17 MuJoCo substeps to match 30Hz, at most 1 ms rounding error",

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    "prefix": "recorded full-scene qpos through branch point; not a free-running prefix",
    "restore": "branch qpos; causal backward finite-difference qvel using mj_differentiatePos; unavailable
warm-start/integration state reset",
    "control": "l4D recorded absolute joint targets; grippers multiplied by 0.0475; mapping identical to ABC
env. step single; no task runtime for mugs",
    "random_walk": "around measured branch joint pose; resample uniform +/-0.5 rad target every 0.5 s; rate
limit 0.6 rad/s; grippers uniform [0,1], rate limit 1/s; clip to joint limits",
    "goal": "separate recorded successful source endpoint, rendered identically for all branches; independent
of perturbation outcome, retrospective demonstration goal",
    "label": "same ABC upright/initial-height rule; report measured terminal and last-one-second stability; no
assumed negative labels",
    "split": "by source episode; all paired branches stay together; inherited training membership remains
uncertain",
    "renderer": "MuJoCo Warp, top camera 224x168, 10fps review videos; all classes share renderer",
    "status": "protocol frozen before generated outcomes"
  },
  "validation": {
    "result": "passed",
    "branches": 18,
    "frames_checked": 15006,
    "checks": [
      "all generated completion labels exactly match pinned upstream evaluator",
      "recorded prefixes byte-identical",
      "paired branches start with identical full restored integration state",
      "all generated physics states finite; no MuJoCo warnings",
      "action grippers within [0,1]",
      "source groups do not cross development/test"
    ]
  },
  "outcomes": [
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      "frames": 1034,
      "success": true
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      "mode": "hold",
      "frames": 1034,
      "success": false
    },
    {
      "episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c",
      "mode": "random_walk",
      "frames": 1034,
      "success": false
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      "mode": "replay_control",
      "frames": 819,
      "success": true
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      "frames": 819,
      "success": false
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      "success": false
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      "frames": 794,
      "success": true
    }
  ]
}

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    },
    {
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      "mode": "replay_control",
      "frames": 752,
      "success": true
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      "mode": "hold",
      "frames": 752,
      "success": false
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      "frames": 752,
      "success": false
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      "mode": "replay_control",
      "frames": 888,
      "success": true
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      "frames": 888,
      "success": false
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      "frames": 888,
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      "frames": 715,
      "success": true
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      "mode": "hold",
      "frames": 715,
      "success": false
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    {
      "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a",
      "mode": "random_walk",
      "frames": 715,
      "success": false
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"threshold_results": {

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"positive_class": "successful",
"unit": "episode",
"dev_episodes": 6,
"test_episodes": 12,
"test_successful": 4,
"test_incomplete": 8,
"threshold_rule": "maximize development balanced accuracy; ties minimize false-success predictions, then
favor higher threshold",
"uncertainty": "source-episode cluster bootstrap; all related branches resampled together, frozen
threshold; single-class replicates omitted",
"methods": {
  "DINOv2": {
    "fit": {
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      "score_direction": "higher means successful"
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    "threshold": 0.8599722981452942,
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        "true_success_pred_incomplete": 0,
        "true_success_pred_success": 2
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    },
    "held_out": {
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      "precision_success": 1.0,
      "recall_success": 1.0,
      "recall_incomplete": 1.0,
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        "true_incomplete_pred_success": 0,
        "true_success_pred_incomplete": 0,
        "true_success_pred_success": 4
      }
    },
    "held_out_95ci": {
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        1.0
      ],
      "balanced_accuracy": [
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      ],
      "recall_success": [
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        1.0
      ],
      "recall_incomplete": [
        1.0,
        1.0
      ]
    }
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      "split": "dev",
      "score": 0.9917763471603394,
      "predicted_success": true
    },

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{
  "episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c/hold",
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  "predicted_success": false
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  "split": "dev",
  "score": 0.726413369178772,
  "predicted_success": false
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  "split": "dev",
  "score": 0.9930462837219238,
  "predicted_success": true
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  "score": 0.7093908190727234,
  "predicted_success": false
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  "score": 0.728168249130249,
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  "predicted_success": true
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  "predicted_success": false
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  "score": 0.6626816391944885,
  "predicted_success": false
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  "split": "test",
  "score": 0.9978718757629395,
  "predicted_success": true
},
{
  "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/hold",
  "split": "test",
  "score": 0.7627826929092407,
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},
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  "split": "test",
  "score": 0.6974966526031494,
  "predicted_success": false
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  "split": "test",
  "score": 0.9932936429977417,
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    "score": 0.7322099208831787,
    "predicted_success": false
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    "predicted_success": false
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    "split": "test",
    "score": 0.9833273887634277,
    "predicted_success": true
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    "score": 0.607895016670227,
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    "split": "test",
    "score": 0.583622932434082,
    "predicted_success": false
  }
]
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"DT": {
  "fit": {
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    "score_direction": "higher means successful"
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  "threshold": -8.942145824432373,
  "development": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
    "confusion_matrix": {
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      "true_incomplete_pred_success": 0,
      "true_success_pred_incomplete": 0,
      "true_success_pred_success": 2
    }
  },
  "held_out": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
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    "recall_incomplete": 1.0,
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      "true_success_pred_success": 4
    }
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  "held_out_95ci": {
    "auroc": [

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    1.0,
    1.0
  ],
  "balanced_accuracy": [
    1.0,
    1.0
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
},
"predictions": [
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    "score": -18.49923324584961,
    "predicted_success": false
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    "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/replay_control",
    "split": "test",
    "score": -2.1928205490112305,
    "predicted_success": true
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    "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/hold",
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    "predicted_success": false
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    "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/random_walk",
    "split": "test",
    "score": -22.700782775878906,
    "predicted_success": false
  }
]

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    },
    {
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      "score": -18.502601623535156,
      "predicted_success": false
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      "split": "test",
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      "predicted_success": false
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      "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/replay_control",
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      "score": -1.8384979963302612,
      "predicted_success": true
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      "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/hold",
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      "predicted_success": false
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      "predicted_success": false
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      "split": "test",
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      "predicted_success": false
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      "score": -33.9699821472168,
      "predicted_success": false
    }
  ]
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    "score_direction": "higher means successful"
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    "recall_incomplete": 1.0,

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    }
  },
  "held_out": {
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    "recall_success": 0.0,
    "recall_incomplete": 1.0,
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      "true_incomplete_pred_success": 0,
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      "true_success_pred_success": 0
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  },
  "held_out_95ci": {
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    ],
    "recall_success": [
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    "recall_incomplete": [
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    ]
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  "predictions": [
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    {
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    }
  ],

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{
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  "score": 26.433333333333334,
  "predicted_success": false
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  "predicted_success": false
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  "score": 25.033333333333335,
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  "score": 29.566666666666666,
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  "split": "test",
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}
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    "predicted_success": false
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    "split": "test",
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      "true_success_pred_success": 2
    }
  },
  "held_out": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
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      "true_success_pred_success": 4
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  "held_out_95ci": {
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      1.0
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    "recall_success": [
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      1.0
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    "recall_incomplete": [
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      1.0
    ]
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      "score": -1.5910629034042358,
      "predicted_success": true
    },
    {

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{
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  "score": -5.169395446777344,
  "predicted_success": false
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{
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  "split": "dev",
  "score": -4.953976154327393,
  "predicted_success": false
},
{
  "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/replay_control",
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  "score": -1.4238965511322021,
  "predicted_success": true
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  "split": "test",
  "score": -5.897188663482666,
  "predicted_success": false
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  "score": -6.108412742614746,
  "predicted_success": false
},
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  "split": "test",
  "score": -0.9455946683883667,
  "predicted_success": true
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{
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  "score": -5.358786582946777,
  "predicted_success": false
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  "predicted_success": false
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  "split": "test",
  "score": -1.4018906354904175,
  "predicted_success": true
}
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    },
    {
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      "score": -5.36187744140625,
      "predicted_success": false
    },
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      "split": "test",
      "score": -5.218265533447266,
      "predicted_success": false
    },
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      "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/replay_control",
      "split": "test",
      "score": -2.5138607025146484,
      "predicted_success": true
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    {
      "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/hold",
      "split": "test",
      "score": -5.818654537200928,
      "predicted_success": false
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      "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/random_walk",
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      "score": -6.158592700958252,
      "predicted_success": false
    }
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"s3": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
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      "true_success_pred_success": 2
    }
  },
  "held_out": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
    "confusion_matrix": {
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      "true_success_pred_incomplete": 0,
      "true_success_pred_success": 4
    }
  },
  "held_out_95ci": {
    "auroc": [
      1.0,

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    1.0
  ],
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    1.0,
    1.0
  ],
  "recall_success": [
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    1.0
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  "recall_incomplete": [
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    1.0
  ]
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    "split": "dev",
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    "predicted_success": false
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    "predicted_success": false
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    "predicted_success": true
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    "score": -6.119126319885254,
    "predicted_success": false
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    "score": -5.886356830596924,
    "predicted_success": false
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  "score": -0.6890887022018433,
  "predicted_success": true
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  "split": "test",
  "score": -5.3669538497924805,
  "predicted_success": false
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  "score": -5.127117156982422,
  "predicted_success": false
},
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  "split": "test",
  "score": -1.2436871528625488,
  "predicted_success": true
},
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  "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/hold",
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  "score": -5.47298002243042,
  "predicted_success": false
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  "split": "test",
  "score": -4.950935363769531,
  "predicted_success": false
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{
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  "split": "test",
  "score": -2.3856911659240723,
  "predicted_success": true
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  "split": "test",
  "score": -5.943408489227295,
  "predicted_success": false
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  "predicted_success": false
}
]
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{
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    "true_success_pred_incomplete": 0,
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"held_out": {
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  "accuracy": 1.0,
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    "true_success_pred_success": 4
  }
},
"held_out_95ci": {
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    1.0
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    1.0
  ],
  "recall_success": [
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  ],
  "recall_incomplete": [
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    1.0
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    "predicted_success": false
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    "predicted_success": true
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    "split": "dev",
    "score": -5.169167518615723,
    "predicted_success": false
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"score": -4.900981903076172,
"predicted_success": false
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{
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  "split": "test",
  "score": -1.3093634843826294,
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  "score": -6.11904239654541,
  "predicted_success": false
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  "predicted_success": false
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  "predicted_success": false
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  "split": "test",
  "score": -2.3576560020446777,
  "predicted_success": true
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  "split": "test",
  "score": -5.978389739990234,
  "predicted_success": false
}
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    {
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      "predicted_success": false
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    "fit": {
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      "C": 1,
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        "S1"
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        -4.072144329547882
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      "std": [
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        1.9770790831434328
      ],
      "coefficients": [
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        0.8433120505934399
      ],
      "intercept": -1.002082720680716
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    "threshold": 0.448230025164487,
    "development": {
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      "precision_success": 1.0,
      "recall_success": 1.0,
      "recall_incomplete": 1.0,
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        "true_incomplete_pred_success": 0,
        "true_success_pred_incomplete": 0,
        "true_success_pred_success": 2
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    },
    "held_out": {
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      "balanced_accuracy": 1.0,
      "accuracy": 1.0,
      "precision_success": 1.0,
      "recall_success": 1.0,
      "recall_incomplete": 1.0,
      "confusion_matrix": {
        "true_incomplete_pred_incomplete": 8,
        "true_incomplete_pred_success": 0,
        "true_success_pred_incomplete": 0,
        "true_success_pred_success": 4
      }
    },
    "held_out_95ci": {
      "auroc": [
        1.0,
        1.0
      ],
      "balanced_accuracy": [
        1.0,
        1.0
      ],
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1.0,
1.0
],
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1.0,
1.0
]
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"predictions": [
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"split": "dev",
"score": 0.7770704028526265,
"predicted_success": true
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"episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c/hold",
"split": "dev",
"score": 0.10713362507382787,
"predicted_success": false
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{
"episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c/random_walk",
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"predicted_success": false
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{
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"predicted_success": true
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"score": 0.10705157153123669,
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"predicted_success": false
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"predicted_success": true
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{
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"predicted_success": false
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"predicted_success": false
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{
"episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/replay_control",
"split": "test",
"score": 0.8208558860338556,
"predicted_success": true
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{

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    "split": "test",
    "score": 0.10236087429512905,
    "predicted_success": false
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    "split": "test",
    "score": 0.10262056333292122,
    "predicted_success": false
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  {
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    "split": "test",
    "score": 0.7945964608366938,
    "predicted_success": true
  },
  {
    "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/hold",
    "split": "test",
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    "predicted_success": false
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    "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/random_walk",
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    "score": 0.11896467200559853,
    "predicted_success": false
  },
  {
    "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/replay_control",
    "split": "test",
    "score": 0.6955640186445826,
    "predicted_success": true
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    "split": "test",
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    "predicted_success": false
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    "split": "test",
    "score": 0.014530461279977331,
    "predicted_success": false
  }
]
},
"DT + S3": {
  "fit": {
    "type": "development-fitted logistic regression",
    "C": 1,
    "features": [
      "DT",
      "S3"
    ],
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      -12.872935136159262,
      -4.046296815077464
    ],
    "std": [
      7.729570962705651,
      2.010529067225712
    ],
    "coefficients": [
      0.8494757342501382,
      0.8459004066462995
    ],
    "intercept": -1.001770452261802
  }
}

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},
"threshold": 0.4499065998983014,
"development": {
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  "balanced_accuracy": 1.0,
  "accuracy": 1.0,
  "precision_success": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": 1.0,
  "confusion_matrix": {
    "true_incomplete_pred_incomplete": 4,
    "true_incomplete_pred_success": 0,
    "true_success_pred_incomplete": 0,
    "true_success_pred_success": 2
  }
},
"held_out": {
  "auroc": 1.0,
  "balanced_accuracy": 1.0,
  "accuracy": 1.0,
  "precision_success": 1.0,
  "recall_success": 1.0,
  "recall_incomplete": 1.0,
  "confusion_matrix": {
    "true_incomplete_pred_incomplete": 8,
    "true_incomplete_pred_success": 0,
    "true_success_pred_incomplete": 0,
    "true_success_pred_success": 4
  }
},
"held_out_95ci": {
  "auroc": [
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    1.0
  ],
  "balanced_accuracy": [
    1.0,
    1.0
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
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"predictions": [
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    "predicted_success": true
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    "split": "dev",
    "score": 0.10560879231881835,
    "predicted_success": false
  },
  {
    "episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c/random_walk",
    "split": "dev",
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    "predicted_success": false
  },
  {
    "episode_id": "episode_019e3c90-fb67-765e-8e8e-12c56ddda3ba/replay_control",
    "split": "dev",

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{"score": 0.809936273441573,
 "predicted_success": true
},
{
  "episode_id": "episode_019e3c90-fb67-765e-8e8e-12c56ddda3ba/hold",
  "split": "dev",
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  "predicted_success": false
},
{
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  "split": "dev",
  "score": 0.11829523017023617,
  "predicted_success": false
},
{
  "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/replay_control",
  "split": "test",
  "score": 0.7868378974413496,
  "predicted_success": true
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{
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  "score": 0.007087470566342854,
  "predicted_success": false
},
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  "score": 0.054370776010207934,
  "predicted_success": false
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{
  "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/replay_control",
  "split": "test",
  "score": 0.8317584010952102,
  "predicted_success": true
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{
  "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/hold",
  "split": "test",
  "score": 0.10191599255876543,
  "predicted_success": false
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  "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/random_walk",
  "split": "test",
  "score": 0.103012949013444,
  "predicted_success": false
},
{
  "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/replay_control",
  "split": "test",
  "score": 0.8006026090457147,
  "predicted_success": true
},
{
  "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/hold",
  "split": "test",
  "score": 0.03540746203028002,
  "predicted_success": false
},
{
  "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/random_walk",
  "split": "test",
  "score": 0.1309800197336101,
  "predicted_success": false
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    "split": "test",
    "score": 0.08211609220260822,
    "predicted_success": false
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    "split": "test",
    "score": 0.014552358465128977,
    "predicted_success": false
  }
]
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"DT + S5": {
  "fit": {
    "type": "development-fitted logistic regression",
    "C": 1,
    "features": [
      "DT",
      "S5"
    ],
    "mean": [
      -12.872935136159262,
      -3.958727906147639
    ],
    "std": [
      7.729570962705651,
      2.020709138851318
    ],
    "coefficients": [
      0.8489792751217451,
      0.8465976355226331
    ],
    "intercept": -1.0011058467082332
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  "threshold": 0.4502469361783752,
  "development": {
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    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
    "confusion_matrix": {
      "true_incomplete_pred_incomplete": 4,
      "true_incomplete_pred_success": 0,
      "true_success_pred_incomplete": 0,
      "true_success_pred_success": 2
    }
  },
  "held_out": {
    "auroc": 1.0,
    "balanced_accuracy": 1.0,
    "accuracy": 1.0,
    "precision_success": 1.0,
    "recall_success": 1.0,
    "recall_incomplete": 1.0,
    "confusion_matrix": {
      "true_incomplete_pred_incomplete": 8,
      "true_incomplete_pred_success": 0,
      "true_success_pred_incomplete": 0,
      "true_success_pred_success": 4
    }
  }
},

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"held_out_95ci": {
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    1.0,
    1.0
  ],
  "balanced_accuracy": [
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    1.0
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  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    1.0,
    1.0
  ]
},
"predictions": [
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    "split": "dev",
    "score": 0.7827363168356453,
    "predicted_success": true
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  {
    "episode_id": "episode_019e298a-84a1-7557-ae97-93326347b45c/hold",
    "split": "dev",
    "score": 0.10448382056554557,
    "predicted_success": false
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  {
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    "split": "dev",
    "score": 0.08206423456695083,
    "predicted_success": false
  },
  {
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    "split": "dev",
    "score": 0.8093386517734267,
    "predicted_success": true
  },
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    "episode_id": "episode_019e3c90-fb67-765e-8e8e-12c56ddda3ba/hold",
    "split": "dev",
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    "predicted_success": false
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    "predicted_success": false
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    "score": 0.7827775412848426,
    "predicted_success": true
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    "split": "test",
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    "predicted_success": false
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    "episode_id": "episode_019e1e35-af9f-7d66-93b0-478d1c079799/random_walk",
    "split": "test",

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    "score": 0.05875754676090327,
    "predicted_success": false
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  {
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    "split": "test",
    "score": 0.8287179021244911,
    "predicted_success": true
  },
  {
    "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/hold",
    "split": "test",
    "score": 0.10327922896159757,
    "predicted_success": false
  },
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    "episode_id": "episode_019e2d91-6e08-7f79-a3e4-c843c73a3ef2/random_walk",
    "split": "test",
    "score": 0.101215801756453,
    "predicted_success": false
  },
  {
    "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/replay_control",
    "split": "test",
    "score": 0.7942614284182674,
    "predicted_success": true
  },
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    "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/hold",
    "split": "test",
    "score": 0.03520728474152032,
    "predicted_success": false
  },
  {
    "episode_id": "episode_019e3c25-a92e-7824-a264-f1da48b9bd3d/random_walk",
    "split": "test",
    "score": 0.12546969833043714,
    "predicted_success": false
  },
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    "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/replay_control",
    "split": "test",
    "score": 0.6962557246300708,
    "predicted_success": true
  },
  {
    "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/hold",
    "split": "test",
    "score": 0.07864577797400821,
    "predicted_success": false
  },
  {
    "episode_id": "episode_019e2261-5abe-7ece-910d-8988dd12106a/random_walk",
    "split": "test",
    "score": 0.016559187171322123,
    "predicted_success": false
  }
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},
"Always successful": {
  "fit": {
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  },
  "threshold": 0.5,
  "development": {
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    "balanced_accuracy": 0.5,
    "accuracy": 0.3333333333333333,
    "precision_success": 0.3333333333333333,
    "recall_success": 1.0,

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"recall_incomplete": 0.0,
"confusion_matrix": {
  "true_incomplete_pred_incomplete": 0,
  "true_incomplete_pred_success": 4,
  "true_success_pred_incomplete": 0,
  "true_success_pred_success": 2
}
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"held_out": {
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  "balanced_accuracy": 0.5,
  "accuracy": 0.3333333333333333,
  "precision_success": 0.3333333333333333,
  "recall_success": 1.0,
  "recall_incomplete": 0.0,
  "confusion_matrix": {
    "true_incomplete_pred_incomplete": 0,
    "true_incomplete_pred_success": 8,
    "true_success_pred_incomplete": 0,
    "true_success_pred_success": 4
  }
},
"held_out_95ci": {
  "auroc": [
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    0.5
  ],
  "balanced_accuracy": [
    0.5,
    0.5
  ],
  "recall_success": [
    1.0,
    1.0
  ],
  "recall_incomplete": [
    0.0,
    0.0
  ]
},
"predictions": [
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    "split": "dev",
    "score": 1.0,
    "predicted_success": true
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    "Synthetic stalls and bounded random-action branches; not natural policy failures.",
    "Arm pose correlates with outcome; need parked-arm failures and completed-task motion controls.",
    "Independent demonstration goals remain retrospective and may closely match successful endpoints.",
    "All scored images re-rendered consistently, with illumination differing from original recordings.",
    "No sample-level training-overlap guarantee.",
    "No threshold tuning on held-out data; adding controls later requires a newly frozen protocol."
]
},
"limitations": "Six source scenes only, with two for development and four for testing. These are synthetic
stalls/action perturbations, not natural policy failures. Arm pose and movement can correlate with the labels;
parked-arm failures and completed-task arm-motion controls are still needed. Matched goals are retrospective
successful demonstration endpoints. Re-rendered illumination differs from original recordings, but all scored
classes and goals use the same renderer. No claim of zero-shot generalization."
}

```